

Khwaja Yunus Ali University

School of Science and Engineering
Department of Computer Science and Engineering

Final Examination: Fall – 2020

Program: B.Sc. in CSE (Batch – 10th) 1st Year 2nd Semester

Course Title: Physics-II

Course Code: PHY 1201

Time: 10 Minutes

Total Marks: 10 X 1 = 10

Student ID:	Batch:	Session:
--------------------	---------------	-----------------

Part 1: MCQ

[These are single best answer type of questions. Each of the following question or statement has five suggested answers or completions. Write down the most appropriate one on your answer script.]

- When the crest of two coherent light waves overlaps at the same time, they will produce
 - the diffraction of light.
 - the resonance.
 - the polarization of light.
 - the constructive interference.
 - the destructive interference.
- What is the phase difference between two waves in a wave front?
 - 0
 - $\pi/4$
 - $\pi/3$
 - $\pi/2$
 - π
- What is the unit of the resolving power of grating?
 - W
 - J
 - kg
 - sec
 - Unitless
- To reduce the glare effect of sunglass, is used.
 - interference
 - diffraction
 - polarization
 - superposition
 - reflection
- A diffraction grating, which contain about 10,000 lines per centimeter, what is the grating constant of it?
 - 10 cm
 - 100 cm
 - 0.001 cm
 - 0.0001 cm
 - 1 cm

P.T.O.

6. Stopping potential increases with the increase of the
- a) photo currents.
 - b) frequency of the incident photon.
 - c) wavelength of the incident photon.
 - d) intensity of the incident light.
 - e) work function of the metal.
7. Half-life of a radioactive substance is 10 days. How long will it take to decrease to 25% of it?
- a) 5 days
 - b) 15 days
 - c) 20 days
 - d) 25 days
 - e) 30 days
8. In sense of relativity, the length of a rigid body in the direction of its motion with uniform velocity v is reduced by a factor in the direction of motion. [Here, c be the light speed]
- a) v/c
 - b) $\ln(v/c)$
 - c) v^2/c^2
 - d) $\sqrt{v^2 - c^2}$
 - e) $\sqrt{1 - (v^2/c^2)}$
9. For a very short velocity of a particle compared to the light speed, which one is the correct relation between rest mass, m_0 and moving mass, m of that particle?
- a) $m_0 > m$
 - b) $m_0 \gg m$
 - c) $m_0 = m$
 - d) $m_0 < m$
 - e) $m_0 \ll m$
10. What kind of change occurs in a nucleus when positron emission occurs?
- a) Proton decreases by one unit and mass numbers remain the same.
 - b) Proton increases by one unit and mass numbers remain the same.
 - c) Neutron decreases by one unit and mass numbers remain the same.
 - d) Neutron and mass number both increases.
 - e) Proton and mass numbers both remain the same.

Khwaja Yunus Ali University

School of Science and Engineering
Department of Computer Science and Engineering

Final Examination: Fall – 2020

Program: B.Sc. in CSE (Batch – 10th) 1st Year 2nd Semester

Course Title: Physics-II

Course Code: PHY 1201

Time: 1 Hour and 50 Minutes

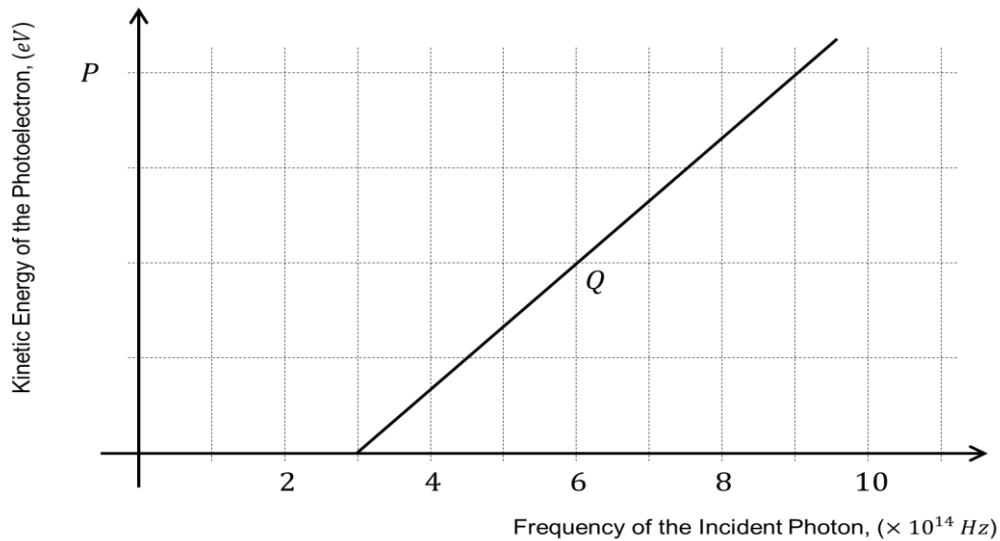
Total Marks: 5 X 6 = 30

Part 2: SAQ

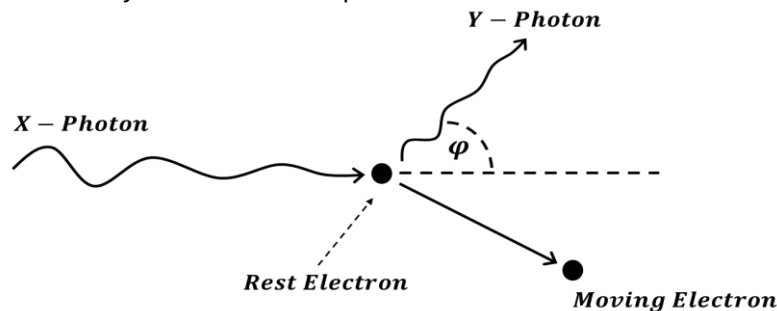
[Answer the question no. 1 and any four (04) from the rest]
[The figures in the right margin indicate full marks for the respective question]

1. See the graph properly and answer the questions that follows:

Kinetic Energy of the Photoelectron vs Frequency of the Incident Photon Graph for a Metal



- a) Define photo current. 1
- b) Write the value of the threshold frequency and find the value of P. 2
- c) Is it possible to produce the photoelectricity from the metal by using the IR of $1.5 \mu\text{m}$ wavelength? – Justify your answer. 3
- 2 a) What is the activity of the radioactive substances? 1
- b) What will happen when ${}^{238}_{92}\text{U}$ emits an α -particle? 2
- c) Find the activity of 1.00 mg of radon, ${}^{222}\text{Rn}$, whose atomic mass is 222 U. 3
3. See the diagram carefully and answer the question that follows:



- a) Give the name of the above event. 1
- b) Compare the properties of X and Y – Photons. 2
- c) If the frequency of X – Photon is $3 \times 10^{19} \text{ Hz}$ and the value of ϕ of the above diagram is 90° . Find the frequency of Y – Photon. 3

P.T.O.

- | | | | |
|----|----|---|---|
| 4. | a) | Distinguish the inertial and non-inertial frame of references. | 3 |
| | b) | At what speed will the mass of the electron be equal to the mass of the proton? | 3 |
| | | | |
| 5. | a) | Define the pointing vector. | 1 |
| | b) | What are the conditions of interference of light? | 2 |
| | c) | In a Young's Double-Slit experiment, the separation between the sources is 0.18mm. the fringes are observed on a screen 90cm away. If with certain monochromatic source of light, the third bright fringe is situated at a distance of 8.1mm from the central bright fringe, find the wave length of light. | 3 |
| | | | |
| 6. | a) | What is grating constant? | 1 |
| | b) | Calculate the angle between the lamp filament and its first diffracted image produced by a fabric with 160 threads/cm. ($\lambda = 6000\text{\AA}$.) | 3 |
| | c) | Write a short note on "Double Refraction of Light". | 2 |

-: The End :-